IRON ORE POTENTIAL

A geological assessment of current districts and future exploration targets across Great Britain, Northern Ireland and Ireland

DESKTOP RESEARCH REPORT
Geology | historical production | prospectivity | exploration strategy

Prepared: 17 October 2026 Evidence base: BGS, GSNI / BGS Earthwise, Historic England, Geological Survey Ireland and public trade data

Principal conclusion: the British Isles contain well-documented iron mineralisation, but the reviewed evidence does not identify a currently operating commercial iron-ore mine or a published modern reserve capable of competing with imported ore. The most meaningful exploration work would test concealed extensions of known high-grade haematite systems, not assume that historic ironstone fields are economic reserves.

Important limitation: this is a regional desktop prospectivity study. It is not a mineral resource or reserve estimate, technical report under JORC / NI 43-101 / PERC, planning opinion, or investment recommendation. All target areas require site-specific data, access rights, environmental screening, sampling and drilling before any resource or economic claim.

Executive summary

The British Isles have a diverse iron-ore record: low-grade Jurassic oolitic ironstones in England, high-grade fault-controlled haematite replacements in western Cumbria and parts of Wales and western England, lateritic iron horizons in the Antrim basalts, blackband / sideritic ironstones associated with coal-bearing strata, and Precambrian banded iron formation in north-west Scotland. The geological styles are real and mapped; the remaining tonnage, grade continuity and recoverability are much less certain. [1-9,12]

The central commercial finding is negative but useful: historical output and a mapped iron-bearing formation do not equal a modern economic deposit. British Jurassic ironstones commonly contain about 17-35% Fe in the Frodingham field and locally only 20-30% Fe in other horizons; they lost out to higher-grade imports. West Cumbrian haematite was substantially higher grade, commonly about 50-60% Fe with low phosphorus, but large-scale mining ended around 1980 and the small-scale Florence operation closed in 2006/07. [1-4,6]

No active commercial iron-ore mine was identified in the current-source review. Historic England records the closure of Florence Mine after a small-scale pigment operation; surviving surface workings and remaining surface ore are heritage / pigment assets, not evidence of a current ironmaking mine. [4,5]

The most defensible exploration thesis is to seek overlooked or concealed extensions of known high-grade, structurally controlled haematite in western Cumbria and analogous Carboniferous Limestone / fault systems in South Wales, the Mendips, Forest of Dean and north Wales. This remains high geological-interest but low-confidence resource exploration: historical orebodies can be irregular, depleted, flooded or protected. Gairloch's magnetite-quartz banded iron formation is a legitimate geological target, but no modern economic tonnage or saleable concentrate quality was identified in the public evidence reviewed. [3,7,8,12]

Antrim's lateritic iron horizons prove that substantial historical iron mining took place in Northern Ireland. The best seams were locally high in iron oxides but thin, weathering-profile bound and ultimately uncompetitive against large high-grade global deposits. Remaining continuity and tonnage would need to be demonstrated rather than inferred from old workings. [9,10]

Decision summary

Question	Finding	Implication
Are iron deposits present?	Yes. Several distinct styles are mapped and historically mined.	Prospectivity is real, but varies strongly by deposit type.
Is there current mining?	No active commercial iron-ore mine was identified in reviewed public sources; Florence closed in 2006/07.	Treat the market as import-dependent unless a current licence/operator is independently verified.
Best geological upside?	Concealed extensions of high-grade replacement haematite; secondarily Gairloch BIF.	Target with structural geology, geophysics, mine plans and borehole data.
Best mapped volume?	Jurassic ironstone systems around Scunthorpe, Cleveland and the Midlands.	Large known formations, but low grade / beneficiation burden limits prospects.
Near-term commercial case?	Weak on current evidence. No modern resource or feasibility case established.	Begin with low-cost data compilation and metallurgical tests, not a mine build.

Report structure

1. Scope and methodology; 2. Deposit models; 3. England's Jurassic ironstones; 4. West Cumbria haematite; 5. Wales and western England; 6. Scotland; 7. Northern Ireland and Ireland; 8. Prospectivity ranking; 9. staged exploration programme; 10. economics, environment and planning; 11. conclusions; references and data caveats.

1. Scope, definitions and current position

Scope. “British Isles” is used geographically to include Great Britain, Northern Ireland and the island of Ireland. The report concentrates on iron minerals potentially relevant to ironmaking: haematite, magnetite, siderite, berthierine / chamosite-rich oolitic ironstone and iron-rich lateritic material. It also notes bog iron and small blackband ironstone occurrences where relevant. This is a screening report, not an exhaustive catalogue of every iron-stained outcrop.

Terminology. An occurrence is evidence of mineralisation; a prospect has a reason to be explored; a mineral resource requires sufficient geological evidence and a defined estimation basis; a reserve requires an economically mineable part of a resource after relevant modifying factors. Historic production, old reserve statements and the existence of an iron-bearing bed are not substitutes for a current resource estimate. BGS’s own mineral-resource mapping warns that mapped mineral resource data can be based primarily on geology with limited economic assessment. [13,15]

Current mining status. Historic England reports that Florence Mine near Egremont was closed by British Steel in 1980 and then partly reopened on a small scale until 2006, principally producing pigment and specialist material; other mine-history sources date final closure to 2007 after pumping ceased and workings flooded. The surviving pithead is now a heritage / arts site. The differences in recorded final dates likely reflect the end of underground production versus the later wind-down or site closure. No active commercial iron-ore extraction operation was found in the current review. [4,5]

Trade context. The UK Parliament published HMRC iron-ore import entries totalling 7.264 million tonnes in 2023, 3.410 million tonnes in 2024 and 2.863 million tonnes in 2025. The 2025 year is explicitly provisional and the figures depend on the listed HS codes and trade declaration basis. They should not be treated as a complete measure of iron units consumed by UK steelmaking without reconciling pellets, agglomerates, scrap, other feedstocks and changes in steelmaking routes. Nevertheless, the data confirms continuing import flows at material scale. [14]

Calendar year	Reported iron-ore imports (Mt)	Status / note
2023	7.264	HMRC entries under codes listed in parliamentary answer.
2024	3.410	Same basis; marked year-by-year by origin.
2025	2.863	Provisional / open year as at the 2026 answer.

Reported UK iron-ore import entries, not a forecast or a full steel-industry iron balance. Source: UK Parliament / HMRC, answer dated 9 April 2026. [14]

Method used in this report

- Geological evidence: mapped host formations, known ore minerals, stratigraphic continuity, structural controls and documented historic mines.
- Resource confidence: whether a modern tonnage / grade estimate or contemporary drilling evidence exists in the reviewed public domain.
- Commercial screening: indicative grade, likely beneficiation, continuity, depth, transport, water / environmental constraints, land use and competition from imported ore.
- Target ranking: expert judgement for early-stage screening, not a statistical mineral prospectivity model. Scores are relative and intentionally conservative.

2. Geological models: where iron ore could occur

The geological model determines what a useful target looks like. The best historical British ores formed in four different ways; each has different exploration signatures and economic limits. The following diagrams are conceptual, not geological cross-sections of specific sites.

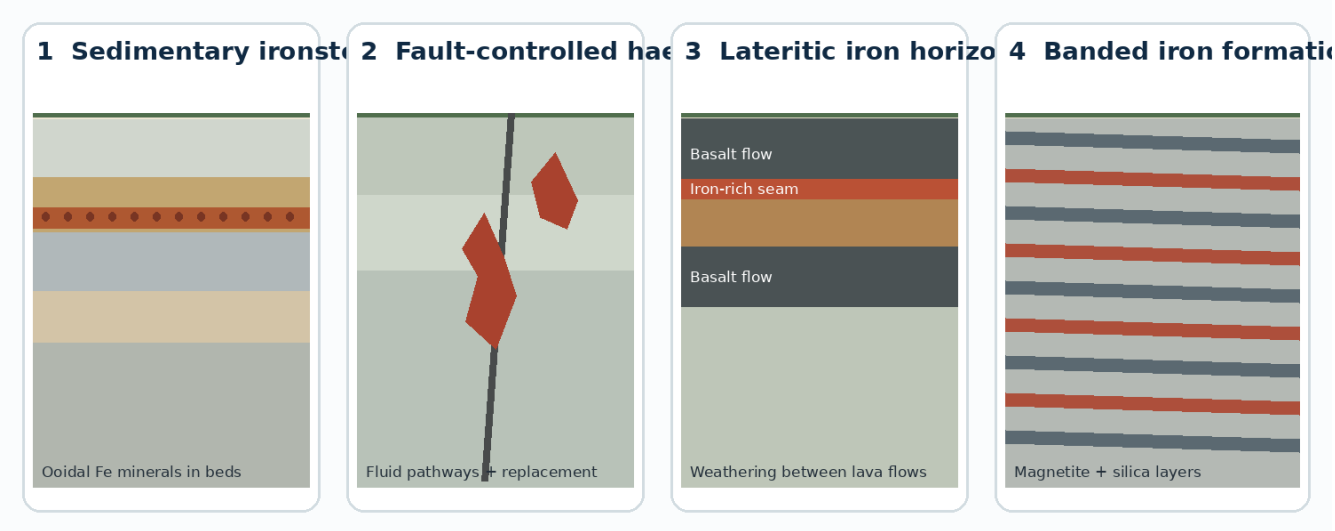


Figure 1. Simplified illustrations of four iron-deposit styles. No scale implied.

A. Sedimentary oolitic ironstone

Iron-rich ooids and iron silicates / carbonates accumulated in Jurassic shallow-marine environments. The beds may be laterally persistent and relatively easy to map, but iron grade varies vertically and laterally; silica, carbonate, phosphorus and clay can dilute the iron units. The Frodingham and Cleveland ironstones are the principal examples. Their historical importance is not proof of modern competitiveness. [1,2,6]

B. Fault-controlled haematite replacement

Iron-bearing fluids moved through faults, fractures and permeable carbonate rocks, replacing limestone with haematite. Orebodies can be high grade and low in phosphorus, but their geometry is irregular: pipes, pods, flat bodies or vein-like replacements rather than a simple continuous sheet. Exploration needs a model of the fault network, carbonate host, paleofluid pathways and post-mineralisation displacement. Western Cumbria is the best-known example; similar styles occur in Wales, the Mendips and the Forest of Dean. [3,7,8]

C. Lateritic iron horizons between basalt flows

Subtropical weathering of basalt concentrated iron and aluminium into a weathering profile between successive lava flows. In County Antrim the iron-rich horizon occurs in the Interbasaltic Formation and may sit above bauxite. The ore can be subhorizontal but thin and locally discontinuous; old mine workings and spoil can obscure the distinction between remaining ore and previously extracted material. [9,10]

D. Banded iron formation (BIF)

Precambrian marine chemical sediments precipitated iron minerals and silica in alternating bands. Metamorphism can recrystallise the original layers into magnetite-rich and quartz-rich bands. Gairloch in north-west Scotland includes oxide-facies iron formation. The key unanswered questions for any commercial target are scale, continuity, average Fe grade, grindability, silica liberation and concentrate recovery, not whether magnetite exists. [12]

3. England: Jurassic ironstone districts

The most extensive mapped iron-bearing formations in England are bedded Jurassic ironstones. Their continuity is geologically better constrained than many replacement deposits, but most are low-grade by modern seaborne iron-ore standards. BGS identifies Frodingham Ironstone, Cleveland Ironstone, Marlstone Rock, Northampton Sand Ironstone and the Dogger / Rosedale magnetic ore among the economically exploited horizons; several other horizons approached ore grade only locally. [1]

Scunthorpe / Frodingham Ironstone - strongest known sedimentary system

The Frodingham Ironstone Member sits in the upper part of the Lower Jurassic Scunthorpe Mudstones Formation. It is a chamositic and sideritic oolite with shell material. BGS describes an average worked thickness of about 9 m, up to around 10 m in parts of the field, with average iron content broadly 17-35%. The bed is thickest in the Scunthorpe area and extends eastwards beneath the Chalk toward Immingham and, in historic borehole interpretations, several miles offshore. Mining ceased in December 1988 as high-grade imported ore and changes in steelmaking economics displaced the local ore. [1,2]

Prospective question: are there low-cost, shallow, accessible blocks or secondary materials that could be upgraded for a specific local process? The mapped continuation east of Scunthorpe is not automatically an economic exploration target: depth, overburden, land use, groundwater, coast / offshore constraints and the burden of beneficiation may dominate. Any re-examination should start by digitising historical boreholes, mine plans, assays and chemical analyses, followed by a modern washability / magnetic-separation test programme.

Cleveland Basin - Eston, Guisborough, Staithes and the North Yorkshire Moors

The Cleveland Ironstone Formation comprises mudstone, silty mudstone and sandstone with sideritic and berthierine-rich ooidal seams. BGS records a formation commonly 20-35 m thick in the basin, thinning markedly toward its margins; named seams typically range from roughly 0.3-1 m, with a Main Seam that was historically the principal ore. The historical main seam reached about 30% Fe in parts of the field. [1,6]

The field has strong historical and geological evidence but a weak near-term bulk-ore case. Better grade sections were mined over a long period; the remaining resource is not defined by modern drilling; historic mine-water discharge is an environmental legacy, and old underground workings raise subsidence and water-management risks. [1,6]

Northampton Sand Ironstone - Northamptonshire and adjoining Midlands

The Northampton Sand Formation was extensively worked in Northamptonshire. Its ironstone locally achieved grades suitable for historic working, but quality varies substantially and neighbouring areas can be much poorer. The BGS memoir for the Melton Mowbray district reports local trial pits averaging only around 20% Fe, while the Leicester memoir describes a Marlstone Rock resource of variable grade, with selected worked intervals in some areas around 25-33% Fe. [16,18]

Marlstone Rock, Claxby and minor horizons

Marlstone Rock Formation is a ferruginous, shelly and ooidal limestone; where iron-rich it weathers to limonitic ironstone. It was historically extracted in the East Midlands, Oxfordshire and near Banbury. BGS regional sources report the end of commercial working as higher-grade imported ironstone became available; the geological formation remains, but its modern value is more likely to be aggregate / building stone in local contexts than a standalone iron-ore operation. [1,6,17]

4. West Cumbria: high-grade haematite and concealed extensions

West Cumbria is the most compelling British Isles example of historically high-grade iron ore. Orebodies occurred between the Lamplugh area and Calderbridge, especially around Egremont, Cleator Moor, Frizington, Beckermet, Florence, Haile Moor and associated workings. BGS describes haematite with roughly 50-60% Fe and very low phosphorus (reported below 0.02%), making it attractive for high-quality steel in the Bessemer era. [3,4]

Deposit geometry and geological controls

The primary targets were massive haematite replacement bodies in Carboniferous Limestone, commonly along faults and joints, with smaller vein deposits in other rocks. BGS notes that ore could range from patchy alteration along fractures to irregular or relatively flat bodies hundreds of metres across. The controlling relationship is not simply “iron-rich limestone”: the best targets would be intersections between structures, permeable carbonate units, suitable cap / seal geometries and the margins of Permo-Triassic basins where iron-bearing fluids could circulate. [3,7,8]

What is left to find?

The historical production demonstrates that very high-grade ore existed, but it also means the most accessible and obvious bodies have largely been mined. A BGS district memoir records that large-scale mining ended with the closure of the Beckermet–Florence mine in the late 1970s and cites a historical estimate of about 1 million tonnes of reserves when large-scale mining ceased, with some material difficult to access. Historic England records that the Florence mine was not worked out when it closed, but later workings stopped and flooded; this does not mean a modern accessible reserve exists. [3,4,5]

High-priority exploration hypotheses

- Concealed carbonate beneath superficial cover: use historical boreholes, mine sections and modern BGS 3D geology to locate untested carbonate blocks adjacent to known ore trends.
- Fault intersections and offsets: reinterpret mine plans and structural lineaments to test whether significant mineralised structures continue below unworked ground or are displaced from old workings.
- Peripheral targets: assess extensions toward Lamplugh, Calderbridge and along faulted carbonate margins only where archival evidence and geophysics agree. Do not assume historic field boundaries define the geological limit.
- Ore quality: assay Fe, P, S, silica, alumina and deleterious trace elements; high Fe alone is insufficient for a modern product specification.

Recommended methods

Start with archival synthesis and a 3D structural model; then use high-resolution ground magnetic and gravity surveys where they can discriminate dense / magnetic ore from host rock, supported by electrical methods or downhole geophysics. Any anomaly should be tested by carefully permitted drilling. Haematite is not always strongly magnetic, so magnetic surveys alone cannot rule targets in or out. Oriented core, density measurements, mineralogy and geochemical assays are essential. Legacy mine workings, unstable ground, flooding, protected geology, land ownership and water quality should be screened before field work.

Prospectivity judgement

Geological potential: moderate-to-high for small concealed high-grade bodies, because the mineralising system is proven. Confidence in a large, continuous, unmined resource: low. Near-term commercial potential: low until drilling defines grade, tonnage and access. The best investment is a low-cost target-generation programme, not a mine-development assumption.

5. Wales, the Mendips and the Forest of Dean

The south and west of Britain contain additional haematite replacement fields with a geological relationship similar to western Cumbria: Carboniferous Limestone host rocks, faults / joints and fluid-driven replacement. These are useful analogues for targeting, but most historical workings were small or have been exhausted, and current resources are not established. [7,8,17]

Vale of Glamorgan / Llanharry

Llanharry was the best-known Welsh iron-ore mine, developed within Carboniferous limestone. The Vale of Glamorgan contains the best-known Welsh iron-ore workings, including Llanharry. BGS geological work discusses competing interpretations of ore genesis but recognises an important hydrothermal / post-Triassic mineralising component. [8,18]

Exploration concept: the target is not the old mine footprint but any untested carbonate compartments along favourable faults and basin margins, under superficial cover or concealed by younger rocks. Start with BGS memoirs, mining plans and borehole records, then integrate gravity / magnetic and structural mapping. The public evidence reviewed did not establish a quantified remaining reserve or a live exploration licence for a particular target.

Forest of Dean

The principal ores were brown haematites (hydrated ferric oxides including goethite), occurring as irregular pockets, lodes and veins that partly replace Carboniferous Limestone. BGS records a very wide historic iron content range (approximately 15-65%), a sign of strong local variability rather than a uniform resource. Ancient scowles and later underground workings document extensive extraction; historic sources attribute the decline to approaching exhaustion of the orefield. [7]

The Forest of Dean is more promising as a structural-geology case study than as a proven new mine area. Exploration would have to show that ore bodies of sufficient size remain outside mined ground and that environmental / land-use conditions allow investigation. Surface iron staining by itself is not a reliable indicator of grade or tonnage.

Mendip Hills

Iron oxides and hydrated iron ores occur in pipes, fissures and replacement patches in Carboniferous Limestone and Dolomitic Conglomerate. Historic production was usually small; the BGS Wells and Cheddar memoir characterises the occurrences as scattered and unlikely to have exceeded hundreds of tonnes annually. These occurrences are more relevant as evidence of the regional fluid system than as independent bulk-ore targets. [17]

North Wales: Prestatyn to Abergele and adjacent areas

North Wales belongs to the broader western British haematite province, where iron mineralisation is known in and around carbonate rocks and structures near the margins of Permo-Triassic basins. At regional scale this provides a rationale to re-examine concealed Carboniferous Limestone and fault intersections, but this review found insufficient public modern data to claim a specific new deposit, grade or tonnage. The area should rank behind West Cumbria for target generation unless historical mine / borehole data reveal an unusually strong untested anomaly. [7,8]

Wales and western England offer moderate geological prospectivity but low resource confidence. The most efficient work is a cross-regional data audit to detect unmined carbonate blocks and structural corridors, rather than independent speculative surveys over each historical district. Avoid confusing heritage / geological sites with permission to explore or mine.

6. Scotland: Gairloch BIF and coalfield ironstone

Scotland presents two fundamentally different iron opportunities. North-west Scotland includes Precambrian iron-formation facies within the Loch Maree Group around Gairloch. The Central Belt and some other coal-bearing districts contain historic sideritic / blackband ironstones. The first is a geological exploration concept with unproven scale; the second is a low-grade legacy resource. [9,12,19]

Gairloch / Flowerdale - magnetite-quartz iron formation

The Gairloch iron formation is part of the ancient Lewisian / Loch Maree Group supracrustal succession. Published mineralogical studies describe oxide facies made of magnetite and quartz, and silicate facies involving minerals such as garnet and amphibole. The Scottish Geological Conservation Review describes finely interbanded quartzite and magnetite-rich layers at Flowerdale. These features confirm a real BIF-type geological system. [12,19]

Why it matters: unlike low-grade Jurassic ironstone, magnetite can in principle be upgraded by grinding and magnetic separation into a higher-grade concentrate. But a BIF outcrop is not automatically a mine. The public sources reviewed do not provide a current modern resource, continuous grade-thickness model, recoverable tonnage or economic study for a Gairloch iron-ore project. Metamorphic complexity may create variable silica liberation and iron distribution; remote logistics, terrain, ecology and potential geological-conservation constraints are significant.

Gairloch exploration questions

- Map the full strike length and structural repetition / disruption of oxide-facies bands using detailed geology and historical geophysical data.
- Use ground magnetics and gravity to model concealed continuation, but calibrate the anomaly response with measured magnetic susceptibility and density from outcrop / core.
- Collect representative channel / core samples across multiple bands, not just visually rich magnetite; measure Fe, SiO2, Al2O3, P, S and trace elements.
- Complete grinding, liberation and Davis-tube / magnetic-separation tests before any claim that a saleable concentrate is realistic.
- Screen conservation designations, land access, water, road / port access and community impacts at the start.

Central Belt, Ayrshire and Scottish coalfield ironstones

Siderite-rich nodules, clay ironstone and blackband ironstone historically supported local ironmaking in Scotland and England. They are typically thin, interbedded with mudstone / coal, and often require mining and processing of substantial non-ore material. This report did not identify a modern Scottish resource or reserve statement supporting renewed bulk production. For a new project these are low priority unless they are a by-product of an existing permitted mine, waste stream, or specialised local metallurgy project. [9,19]

Scotland prospectivity judgement

Gairloch is the more interesting geological exploration target because a magnetite-bearing formation is documented, but it remains high-risk and low-confidence in commercial terms. Coalfield ironstone is lower priority. Neither should be described as a proven reserve.

7. Northern Ireland and Ireland

Northern Ireland: Antrim lateritic iron ores

County Antrim hosted substantial historic iron mining, especially around Glenravel, Cargan, Parkmore, Newtowncrommelin and Glenariff. The ore occurs in the Palaeogene Interbasaltic Formation between basalt flows, formed by tropical weathering of the Antrim Lava Group. BGS describes a thin higher-grade pisolitic layer, commonly around 30 cm, above a thicker lower-grade “pavement”; the former may contain 30-60% total iron oxides and the latter about 25% iron oxides. Historic extraction is recorded at around 5 million tonnes. [9,10]

The distinction between “iron oxides” and elemental iron matters: 30-60% total iron oxides is not the same statement as 30-60% Fe. The seam thickness, profile geometry, oxidation state, silica / alumina content and continuity all require re-assay. The Antrim district became uncompetitive against large, high-grade global iron ores by the mid-20th century. [9]

Remaining potential in Antrim

Any remaining opportunity would be a small, localised re-evaluation of unmined interbasaltic horizons or selected mine residues, not a proven bulk export proposition. Historic mine plans and spoil heaps could help identify where ore was removed versus left in place, but legacy underground workings, surface instability, protected landscapes, hydrology, bauxite association and community / heritage constraints make field access and permitting central. The Cargan Water / Binvore area is recognised for its geological and mining heritage. [9,10]

Republic of Ireland: ironstone, bog iron and low-grade carbonate ores

The public evidence reviewed points to scattered historical iron occurrences rather than a modern, documented iron-ore resource of international scale. Historical literature records sideritic / concretionary iron in Carboniferous shales, including around the Lough Allen district, and various local haematite, magnetite and bog-iron occurrences. These styles can be geologically interesting but often suffer from low grade, high dilution, small or discontinuous bodies, and limited modern drilling data. This report did not locate a current public feasibility study or modern resource statement for a Republic of Ireland iron-ore project. [20,21]

For any Republic of Ireland target, Geological Survey Ireland’s GeoData, mineral-deposit data, GOLDMINE archive and prospecting-licence records should be queried before field activity. Exploration reports may be withheld from public view for current prospecting areas under the state’s rules, so absence of public reports is not proof that no private exploration is taking place. [11,20]

Island of Ireland prospectivity judgement

Antrim has the best demonstrated iron-ore history on the island but weak modern economics; Ireland’s Republic has lower visible evidence for large-scale iron potential in this desk review. The first step should be data-led: compare existing deposit records, old mine plans, assays and licence coverage, then pursue only targets with evidence of thickness, grade and scale beyond the historic small workings.

8. Comparative prospectivity ranking

The chart below separates strength of geological evidence from commercial outlook. A high geology score means the occurrence / model is well documented, not that the deposit is mineable. Commercial ratings are deliberately low where modern grades, tonnage, recovery, access or permitting are unproven.

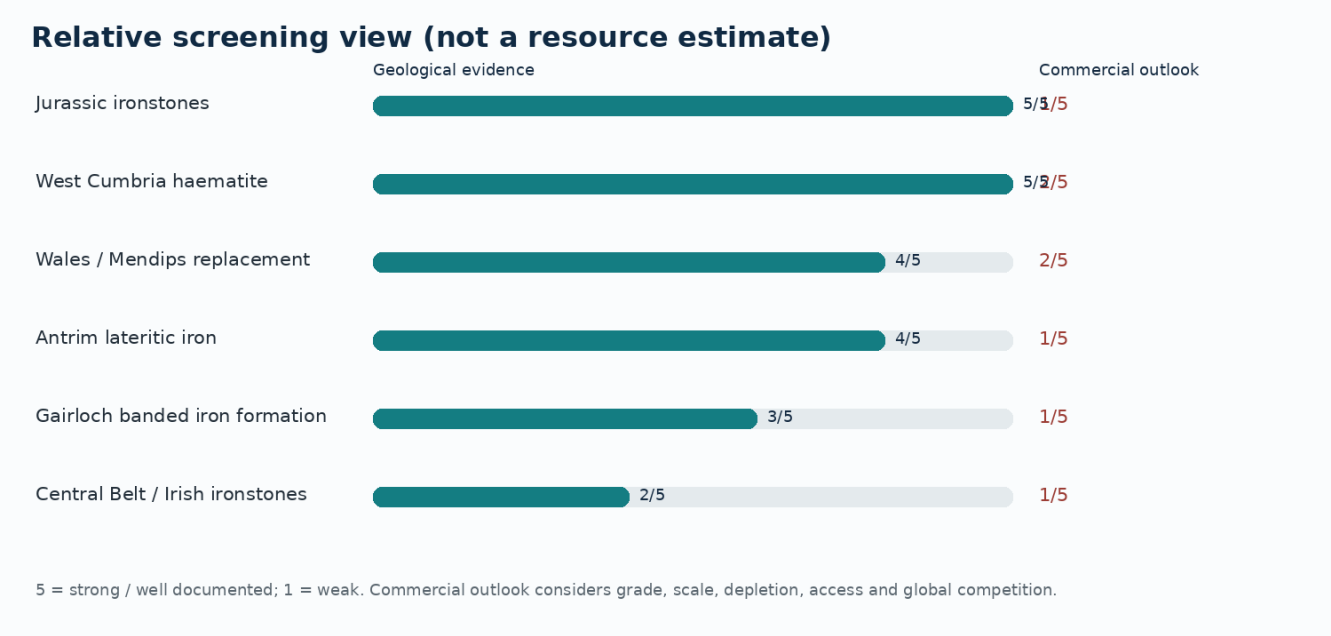


Figure 2. Desktop screening scores (1 = weak; 5 = strong). Commercial outlook is an ordinal judgement, not a price forecast or feasibility result.

Rank / area	Geological evidence	What is known	Main uncertainty / downside	Priority
1. West Cumbria replacement haematite	Very strong historical proof; favourable deposit model.	High-grade haematite in faulted Carboniferous Limestone; major historic production. [3,4]	Known bodies depleted / inaccessible; flooded workings; no modern reserve.	High for data-led target generation; no mining commitment.
2. Jurassic ironstones: Scunthorpe / Cleveland / Midlands	Very strong mapping and historic mining records.	Laterally persistent beds; Frodingham about 17-35% Fe, with other horizons variable. [1,2,6]	Low grade, overburden, beneficiation, older workings and competition from imports.	Medium for metallurgy / by-product study; low for standalone mine.
3. Wales / Mendips / Forest of Dean	Strong evidence of replacement mineralisation.	Irregular haematite pockets / veins in carbonate rocks near structures. [7,8,17]	Highly variable grade / geometry; historical depletion and sensitive land use.	Medium for regional data compilation.
4. Gairloch BIF	Confirmed magnetite-quartz iron formation.	Precambrian iron-formation facies within the Loch Maree Group. [12,19]	No modern tonnage / grade / recovery model found; remote and sensitive setting.	Low-to-medium; geoscience-led only.
5. Antrim lateritic iron	Strong historic evidence; known weathering horizon.	Subhorizontal iron-rich beds; around 5 Mt historical extraction. [9,10]	Thin and variable seams; grade basis must be clarified; global competition.	Low; residues / niche local studies only.
6. Central Belt / Republic ironstones & bog iron	Documented but scattered.	Sideritic / blackband beds, concretions and shallow iron precipitates. [9,20]	Low grade, low continuity, dilution and poor modern resource evidence.	Very low for a standalone bulk mine.

9. Recommended exploration programme

A disciplined programme should use inexpensive existing data to kill weak ideas before field spending. This sequencing is aligned with the BGS guidance that regional exploration requires integration of past mining, known occurrences, geological targeting, exploration methods, mineral rights and planning constraints. [13]

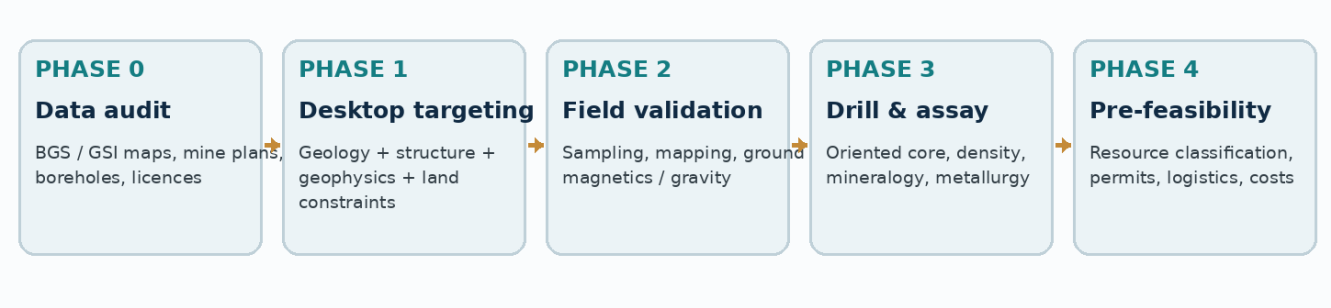


Figure 3. Proposed staged exploration workflow. A phase should not progress unless the evidence clears a defined decision gate.

Phase 0 - Data audit (first gate)

- Compile BGS 1:50k / 1:10k geology, borehole logs, geophysical datasets, historical mineral reports and mine plans. Use GSI / GSNI sources for the island of Ireland.
- Georeference old plans and assay sheets; reconcile obsolete mine names, coordinate systems, seam terminology and grade units.
- Build a GIS inventory separating known mine workings, recorded showings, mapped host formations, protected areas, land ownership, groundwater and access constraints.
- Require a specific geological hypothesis for each target: host, structure / stratigraphy, ore mineral, expected geometry and testable evidence.

Phase 1 - Desktop prospectivity and geophysics

- West Cumbria / Welsh carbonate replacements: structural interpretation, mine-plan gaps, gravity and magnetic data where suited to haematite / associated alteration.
- Gairloch BIF: ground magnetic susceptibility calibration and gravity / magnetic modelling to estimate continuity, with careful controls for metamorphic complexity.
- Jurassic / Antrim stratiform horizons: reconstruct seam thickness, cover, mine depletion and historical assay distributions; check if any plausible product pathway exists before new surveys.
- Rank targets with a transparent decision matrix and pre-defined kill criteria.

Phase 2 - Field validation and sampling

Map bedrock, structure and mine legacy conditions; sample only with permission. Use channel samples or oriented core where geometry matters, with blank / standard / duplicate QA/QC and independent assays. Outcrop grabs are biased and cannot estimate average grade.

Phase 3 - Drilling, metallurgy and resource work

Drill only after a coherent target is established. Collect oriented core, density, geotechnical and hydrogeological data; log and assay continuous intervals rather than cherry-picked sections. Identify the iron minerals and impurities. Test liberation, grinding, separation, mass yield, Fe recovery, concentrate grade, water use and tailings behaviour. At each phase, issue a go / hold / stop memorandum covering data quality, target rationale, access, environmental status, grade-thickness evidence, key uncertainty and next-phase costs. Commission a resource statement only after drilling and sampling support a recognised reporting standard.

10. Economics, planning and environmental constraints

Iron ore is a bulk commodity. A British project must compete not only on geology but against imported ores with established port, rail and processing chains. High-grade historical haematite is attractive in principle, but only if an unmined body has sufficient scale and can be mined legally and safely. Low-grade ironstone requires upgrading; cost and environmental footprint of beneficiation can erase any apparent advantage of local geology.

Product quality and processing questions

Factor	What needs to be measured	Why it matters
Fe grade / mineralogy	Total Fe, Fe-bearing phases, weathering, liberation size.	Grade alone can mislead where iron is locked in silicates / carbonates.
Impurities	SiO2, Al2O3, P, S, Mn, MgO, alkalis and trace elements.	Can affect slag, sintering / pelletising, steel chemistry or customer acceptance.
Recovery & yield	Mass yield, Fe recovery, concentrate grade, fines generation and water use.	A 30% Fe rock can yield a small saleable mass if much is rejected.
Mine geometry	Thickness, dip, continuity, depth, overburden, faults and waste-to-ore ratio.	Determines whether extraction can be continuous and low cost.
Infrastructure	Power, water, rail / road, port distance, plant / tailings sites.	Bulk ores tolerate little expensive haulage or handling.
Environmental liabilities	Acid / ferruginous drainage, mine water, dust, tailings, habitats and heritage.	Historic workings can leave long-term liabilities; Cleveland mine water is a documented example. [6]

Planning, rights and environmental constraints

Mineral extraction usually requires planning permission and depends on land rights, environmental assessment, water management, waste and mine-safety regulation. BGS planning sources stress the role of mineral-resource evidence in land-use decisions. The area-specific screen must include SSSIs / geological conservation sites, National Parks and protected landscapes, nature designations, archaeology, groundwater source protection, flood risk, urban development and local mineral plans. No map-based desktop assessment can replace a formal pre-application review. [13,15]

Legacy mining risks

Old shafts, unrecorded workings, flooded underground mines, unstable tips and contaminated or iron-rich mine water are not just safety issues - they alter the cost and liability profile. For example, Cleveland’s historic workings have produced ferruginous mine-water discharges and some very acidic discharges. Existing spoil can contain recoverable iron in some contexts but may also carry sulphides, fines and contaminants. Any residue-recovery idea needs representative sampling, geotechnical assessment and waste-status review. [6]

Potential commercial routes to examine

- Small high-grade replacement ore: only if drill-defined continuity and access are proved; likely the most geologically attractive route but the least certain at scale.
- Magnetite concentrate from a BIF: only after bulk sampling proves liberation and recovery; the lack of a known resource is the first barrier, not the process design.
- Ironstone beneficiation or residues: a research / pilot-scale opportunity if an existing local industrial customer can use the product and waste can be managed economically; not a default mine case.
- Specialty pigment and mineral products: potentially viable at small scale for selected haematite, but this is a different business from ironmaking ore and cannot support bulk-ore assumptions.

11. Conclusions and recommended target order

The British Isles are geologically capable of hosting iron mineralisation, and historical mining demonstrates this across several distinct deposit types. The issue is no longer whether iron-bearing rocks exist; it is whether a remaining body can deliver enough saleable iron, at reliable quality, with safe access and acceptable environmental impact, to compete with import supply. On current public evidence, no candidate in this review clears that test.

Recommended order for future research

Order	Target	Next best step	Reason to pursue / stop
1	West Cumbria, concealed high-grade haematite extensions	Mine-plan and borehole integration; 3D structural model; targeted geophysics.	Proven high-grade system. Stop quickly where historic depletion, flooding, access or conservation blocks a target.
2	Wales / Mendips / Forest of Dean replacement systems	Regional screening of Carboniferous Limestone near faults and basin margins.	Same broad ore model; lower public confidence in remaining scale. Require untested target evidence.
3	Gairloch BIF	Existing geochemistry + magnetics + representative sampling + concentration tests.	Magnetite-quartz iron formation is proven but no modern resource; conservation and logistics are significant.
4	Jurassic ironstone districts	Historical assay / borehole audit and modern beneficiation tests at a specific site.	Good geological continuity; low grade and historic economic displacement. Only proceed with a distinct processing advantage.
5	Antrim lateritic iron / historical residues	Map mined versus unmined horizons; clarify chemistry and volume.	Historic mining proved the system, but thin beds and global competition limit scale.
6	Coalfield, bog iron and scattered Republic of Ireland occurrences	Use data only to identify by-product / niche cases.	Weak evidence for a stand-alone bulk iron-ore operation.

Recommendation: commission a focused Phase 0 desktop programme for West Cumbria and the wider western British haematite province, with clear stop/go criteria. Do not present any British Isles target as a reserve, and do not model a mine on historic production figures alone.

What would change this conclusion?

A materially stronger case would require: (1) a drill-defined body with several million tonnes or more of coherent mineable geometry, or a compelling niche deposit; (2) demonstrated product quality and metallurgical recoveries; (3) a realistic plan for water, mine safety, waste and environmental liabilities; (4) rights and permission pathway; and (5) a customer / logistics route capable of matching imported ore economics. These are decision criteria, not predictions that such a body exists.

Appendix A. Import data snapshot

The UK Parliament response dated 9 April 2026 provides HMRC-reported iron-ore imports by origin for 2023-2025. It lists major 2025 origins including Sweden (0.910 Mt), Brazil (0.598 Mt), Canada (0.566 Mt), Mauritania (0.356 Mt) and South Africa (0.188 Mt). The 2025 year is open / provisional. The table uses iron-ore commodity codes named in the answer; it is not a full material-flow account for iron and steel. [14]

Origin	2023 (Mt)	2024 (Mt)	2025 (Mt)
Sweden	0.945	0.651	0.910
Brazil	1.293	0.524	0.598
Canada	1.290	0.497	0.566
Norway	1.187	0.369	0.028
United States	0.597	0.492	0.216
Mauritania	0.315	0.249	0.356
South Africa	0.745	0.016	0.188
Grand total	7.264	3.410	2.863

Note: the table reproduces the most relevant visible origins from the parliamentary response; some countries are blank in 2025 where no quantity is shown. The source notes the country-of-origin basis and limitations in declarations through Intrastat. Use the linked original for the full table and precise codes. [14]

Appendix B. Terms and evidence limitations

Term	Meaning in this report
Geological prospectivity	How favourable the mapped geology / deposit model appears for mineralisation. It does not imply grade, tonnage or economics.
Historical resource / reserve	A past estimate made under older assumptions. It may not be current, compliant with modern reporting standards or accessible today.
Grade	Concentration of iron or iron-bearing minerals; “% total iron oxides” should not be confused with elemental % Fe.
Concealed extension	A hypothesised continuation below cover or beside an old mine, requiring geophysical / drilling confirmation.
Resource statement	Requires representative sampling, QA/QC, geological modelling, estimation methods, classification and competent-person review under a recognised standard.

References and key data sources

Sources accessed 17 October 2026 unless a publication date is stated. Inline bracketed numbers refer to the source list below. BGS and government sources are prioritised; historical and specialist sources are used where they provide site-specific information. Links are embedded in source titles.

[1] British Geological Survey, Eastern England from the Tees to The Wash: British regional geology. Ironstone horizons, grades, historical output and remaining bedded formations. Open source
https://webapps.bgs.ac.uk/memoirs/docs/B01344.html

[2] British Geological Survey, Geology of the country around Kingston upon Hull and Brigg. Frodingham Ironstone thickness, mineralogy, extension beneath cover and 1988 production cessation. Open source
https://webapps.bgs.ac.uk/memoirs/docs/B01547.html

[3] British Geological Survey, Geology of the west Cumbria district. Haematite ore styles, host rocks, historical production and structure. The source contains older operating-status text; current closure status is taken from Historic England / mine-history sources. Open source
https://webapps.bgs.ac.uk/Memoirs/docs/B01506.html

[4] Historic England, Florence iron mining pit head, National Heritage List for England, List Entry 1449212. Mine chronology and heritage status. Open source
https://historicengland.org.uk/listing/the-list/list-entry/1449212

[5] Northern Mines Research Society, Florence Mine. History of the small-scale post-1980 operation and final closure in 2007. Open source
https://www.nmrs.org.uk/assets/mines/iron/cumberland/florence.html

[6] British Geological Survey, Northern England British Regional Geology and related Cleveland Basin geology. Cleveland ironstone, grade context and historical iron ore field. Open source
https://webapps.bgs.ac.uk/Memoirs/docs/B06817.html

[7] British Geological Survey, Economic minerals other than coal, Bristol and Gloucester region. Forest of Dean haematite ore style, range of iron contents and historical depletion. Open source
https://earthwise.bgs.ac.uk/index.php/Economic_minerals_other_than_coal,_Bristol_and_Gloucester_region

[8] British Geological Survey, Geology of the South Wales Coalfield memoir. Carboniferous Limestone haematite and discussion of the Llanharry ore genesis. Open source
https://webapps.bgs.ac.uk/memoirs/docs/B01833.html

[9] British Geological Survey / Geological Survey of Northern Ireland, Bulk minerals, mineral resources, Northern Ireland. Antrim lateritic iron ores, blackband ironstone, cherty ironstone and historical extraction. Open source
https://earthwise.bgs.ac.uk/index.php/Bulk_minerals,_mineral_resources,_Northern_Ireland

[10] Department of Agriculture, Environment and Rural Affairs, Northern Ireland, Antrim Coast and Glens AONB; and National Museums NI Earth Science Conservation Review, Cargan Water / Binvore. Mining history and landscape context. Open source
https://www.daera-ni.gov.uk/articles/antrim-coast-and-glens-aonb

[11] Geological Survey Ireland / Department of Climate, Energy and the Environment, Exploration Company Reports. Prospecting licence information and public-report limitations. Open source
https://www.gov.ie/en/department-of-climate-energy-and-the-environment/services/exploration-company-reports/

[12] British Geological Survey, Flowerdale, Geological Conservation Review. Gairloch / Loch Maree Group banded iron formation and magnetite-rich layers. Open source
https://webapps.bgs.ac.uk/Memoirs/docs/GCR34_Flowerdale.html

[13] British Geological Survey, Exploration for Metalliferous and Related Minerals in Britain: A Guide, second edition, created / updated 16 February 2024. Exploration methods, previous work, rights and planning context. Open source
https://www.bgs.ac.uk/mineralsuk/download/exploration-for-metalliferous-and-related-minerals-in-britain-a-guide/

[14] UK Parliament, Written Question HL16023, answer dated 9 April 2026. HMRC Overseas Trade Statistics: UK iron-ore imports by origin for calendar years 2023-2025. 2025 provisional. Open source
https://questions-statements.parliament.uk/written-questions/detail/2026-03-25/hl16023

[15] British Geological Survey, BGS Geology datasets and Onshore maps. Digital geological mapping and mineral-resource map limitations / use. Open source
https://www.bgs.ac.uk/datasets/bgs-geology/

[16] British Geological Survey, Geology of the Melton Mowbray district. Local Northampton Sand Ironstone grade variability and historical working. Open source
https://webapps.bgs.ac.uk/Memoirs/docs/B06079.html

[17] British Geological Survey, Geology of the country around Wells and Cheddar. Mendip iron ore occurrences and historical production. Open source
https://webapps.bgs.ac.uk/memoirs/docs/B01668.html

[18] British Geological Survey, Geology of the Leicester district. Marlstone Rock Formation, ironstone thickness and historical iron content. Open source
https://webapps.bgs.ac.uk/Memoirs/docs/B06819.html

[19] Keele University Research Repository, Mineralogy, petrology and geochemistry of the banded iron formation, Gairloch, N.W. Scotland; NERC Open Research Archive, Mineral exploration in Lewisian supracrustal and basic rocks. Open source
https://keele-repository.worktribe.com/output/415004/mineralogy-petrology-and-geochemistry-of-the-banded-iron-formation-gairloch-nw-scotland

[20] Geological Survey Ireland, GOLDMINE digital archive. Historical maps, mineral-locality reports, mine records and exploration reports. Open source
https://secure.decc.gov.ie/goldmine/index.html

[21] Historical specialist literature on Irish iron ores, including blackband ironstone and bog iron occurrences. Used only to characterise scattered historical occurrences, not to infer a modern resource. Open source
https://www.dmm.org.uk/minequar/4406-01.htm

[22] Younger, P. L. (2002), The importance of pyritic roof strata in aquatic pollutant release from abandoned mines in a major, oolitic, berthierine-chamosite-siderite iron ore field, Cleveland, UK, Geological Society, London, Special Publications. Open source
https://doi.org/10.1144/GSL.SP.2002.198.01.16

Prepared as a regional geology and prospectivity screening report. Geological interpretations are based on public evidence and are subject to revision as new mapping, drilling, assay results, mineral rights, environmental requirements or market conditions change. The report should not be used as a substitute for site-specific technical, legal, environmental or financial due diligence.